In [ ]:
```python
from ultralytics import YOLO   #Pragya Singhal,CU24220050,43
import cv2
import matplotlib.pyplot as plt

def run_object_detection():
    print("Loading YOLOv8 model...")
    model = YOLO("yolov8n.pt")
    print("Model loaded successfully!\n")
#Pragya Singhal,CU24220050,43
    image_path = "street.jpeg"
    print(f"Running object detection on {image_path}...")

    # 3. Perform object detection / inference
    results = model(image_path)

    # 4. Process, display, and save the results   #Pragya Singhal,CU24220050,43
    for r in results:
        annotated_frame = r.plot()

        annotated_rgb = cv2.cvtColor(annotated_frame, cv2.COLOR_BGR2RGB)

        # Display the image  #Pragya Singhal,CU24220050,43
        plt.figure(figsize=(10, 10))
        plt.imshow(annotated_rgb)
        plt.axis("off")
        plt.show()
        output_path = "output_detected.jpg"
        cv2.imwrite(output_path, annotated_frame)
        print(f"Successfully saved output image as '{output_path}'\n")

        print("Detected Objects Summary:")
        for box in r.boxes:
            class_id = int(box.cls[0])
            class_name = model.names[class_id]
            confidence = float(box.conf[0])
            print(f" -> Object: {class_name} | Confidence: {confidence * 100:.2f}%")

if __name__ == "__main__":
    run_object_detection()
```

```
Creating new Ultralytics Settings v0.0.8 file
View Ultralytics Settings with 'yolo settings' or at 'C:\Users\hp\AppData\Roaming\Ultralytics
\settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For
help see https://docs.ultralytics.com/usage/settings.
Loading YOLOv8 model...
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt to 'yolo
v8n.pt': 100% ━━━━━━━━━━━━ 6.2MB 6.7MB/s 0.9s.8s<0.1ss.5s
Model loaded successfully!

Running object detection on street.jpeg...

image 1/1 c:\Users\hp\Pragya-Singhal_CU24220050_Experiment9\street.jpeg: 352x640 3 persons, 6
cars, 1 truck, 3 traffic lights, 217.4ms
Speed: 19.8ms preprocess, 217.4ms inference, 27.6ms postprocess per image at shape (1, 3, 352,
640)
```




```
Successfully saved output image as 'output_detected.jpg'

Detected Objects Summary:
 -> Object: car | Confidence: 88.97%
 -> Object: car | Confidence: 84.49%
 -> Object: person | Confidence: 74.31%
 -> Object: traffic light | Confidence: 68.48%
 -> Object: car | Confidence: 61.32%
 -> Object: car | Confidence: 50.15%
 -> Object: traffic light | Confidence: 44.58%
 -> Object: car | Confidence: 43.08%
 -> Object: person | Confidence: 39.66%
 -> Object: car | Confidence: 32.45%
 -> Object: person | Confidence: 27.18%
 -> Object: truck | Confidence: 26.78%
 -> Object: traffic light | Confidence: 26.43%
```